SACHIIN M

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EDUCATION

B.E. Computer Science & Engineering (AIML), KPR Institute of Engineering and Technology, Coimbatore, India 2023–2027

CGPA: 7.5 / 10

WORK EXPERIENCE

Software Developer Intern @ Zettaone Technologies, Bengaluru Jul – Aug 2025

Flask · React Native · OpenCV · EasyOCR · YOLOv8

- Architected a **PCB component deviation detection** system using **YOLOv8 + EasyOCR**, identifying manufacturing defects in real time via a Flask-based CV pipeline in a production environment.
- Engineered **VISYRE**, a full-stack mobile app (React Native + Flask) for QR/barcode decoding and OCR-based text extraction, eliminating manual data entry across warehouse workflows.
- Built **Climora**, a cross-platform React Native app delivering real-time weather and soil insights via geolocation APIs with live sensor data integration and custom city search.

Research Intern @ Providence University, Taiwan Oct – Nov 2025

CNN · PyTorch · Grad-CAM · Medical Imaging

- Trained a **CNN-based pneumonia classifier** on **7,750 chest X-ray images**, achieving **92.44% overall accuracy**, **0.94 F1-score on pneumonia class**, and **0.93 weighted precision** on a 622-image test set.
- Integrated **Grad-CAM** heatmaps to localise prediction regions and generated **automated PDF medical reports** per patient for interpretable clinical diagnostic output.
- Applied SMOTE, rotation, flipping, zoom, and contrast augmentation to resolve 4:1 class imbalance and improve generalisation across unseen X-ray datasets.

PROJECTS

Morpheus – Natural Language to Blender via MCP In Development

Ollama · MCP · Blender Python API (bpy) · FastAPI · React

- Integrated **Model Context Protocol (MCP)** with Blender to enable LLM-driven scene control – allowing **Ollama** (local LLM) to directly manipulate 3D objects, materials, and lighting through natural language prompts.
- Architected a **custom MCP server** bridging **Ollama** and Blender's Python API (bpy), enabling structured tool calls to create, modify, and render 3D scenes without manual scripting.

VoiceScript – Real-time Speech Transcription System 2025

React · FastAPI · WebSocket · faster-whisper · Silero VAD

- Built a **self-hosted real-time transcription system** using **fasterwhisper** (large-v3-turbo) for accurate speech to text processing, achieving sub-second streaming latency with zero API cost.
- Integrated **Silero VAD** for voice activity detection and **LocalAgreement algorithm** for stable partial transcripts, eliminating word-level jitter across live audio streams.

RESEARCH PUBLICATION & PRESENTATION

Sachiin Megarajan, Khire Rushikesh Ulhas, Ramesh Kumar T, Sutapa Debbarma, *"Pneumonia Detection from Chest X-Ray Images using Dense Net"*, All India Seminar on AI, Design Thinking and Innovation, Institution of Engineers (India), Tripura State Centre Jan 2026

SKILLS

Languages	Python, JavaScript, Java, SQL
Frameworks	React, React Native, FastAPI, Flask, PyTorch, TensorFlow, Angular
ML / CV	OpenCV, YOLOv8, EasyOCR, Scikit-learn, Grad-CAM, faster-whisper, Silero VAD
Tools	Git, Docker, Streamlit, VS Code, Blender (bpy), Ollama
Databases	MySQL, MongoDB